Juniper Core Node BLR-CORE-01 Hardware/Environmental Fault Resolution SOP

Node: BLR-CORE-01 | **Vendor:** Juniper | **Severity:** Minor | **Alarm Code:** ALARM_2730

Description

The Juniper Core Node BLR-CORE-01 hardware/environmental fault, indicated by Alarm Code ALARM_2730 and Severity Minor, is typically a result of ASIC memory parity errors, line card crashes, or power supply redundancy loss. This failure domain occurs due to the inherent complexity of the Juniper MX Series routers, which employ a combination of ASICs and line cards to facilitate high-speed packet processing and forwarding. The MX Series routers utilize ASICs to handle tasks such as packet buffering, caching, and lookup, while line cards are responsible for handling specific network protocols and interfaces. In the event of a hardware or environmental fault, the line card or ASIC may fail, resulting in a loss of redundancy and potentially causing the ALARM_2730 alarm to be triggered. To mitigate this issue, it is essential to promptly identify and address the root cause of the failure, which may involve replacing the faulty line card, repairing or replacing the ASIC, or restoring power supply redundancy.

Pre-requisite Checks

- check: Confirm the site location and dispatch a remote hands team to the Unknown Site if necessary
- check: Verify the redundant link status to ensure optimal network redundancy
- check: Gather relevant logs and alarm history to identify potential patterns or correlations

Potential Root Causes

Cause	Probability
ASIC memory parity errors caused by overheating or voltage fluctuations	High
Line card crashes due to software bugs or hardware malfunctions	Medium
Power supply redundancy loss due to failed or malfunctioning units	Low

Diagnostic Commands & Expected Outputs

```
Command: show chassis hardware, Expected Output: Chassis hardware information, including temperature, voltage, and fan status, Description: This command provides a snapshot of the chassis hardware, including temperature, voltage, and fan status, which can help identify potential environmental or hardware-related issues.

Command: show chassis fpc, Expected Output: FPC status, including FPC number, state, and reason for failure, Description: This command displays the status of the FPCs (Field Programmable Gate Arrays) in the chassis, including FPC number, state, and reason for failure, which can help isolate the issue to a specific FPC.
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```
Command: show log messages | match ALARM_2730, Expected Output: Log messages related to the ALARM_2730 alarm, including timestamp and description, Description: This command filters the system logs to display messages related to the ALARM_2730 alarm, including timestamp and description, which can provide valuable context and help identify the root cause of the issue.
```

Resolution Steps

Step 1: Step: Replace the faulty line card, Command: request chassis fpc replace with , Description: This command initiates the replacement of the faulty line card with a new one, restoring redundancy and resolving the ALARM_2730 alarm.

Step 2: Step: Repair or replace the ASIC, Command: request chassis asic replace with , Description: This command initiates the repair or replacement of the faulty ASIC, which can help resolve the issue and prevent future occurrences.

Step 3: Step: Restore power supply redundancy, Command: request chassis psu replace with , Description: This command initiates the replacement of the failed or malfunctioning PSU with a new one, restoring power supply redundancy and resolving the ALARM_2730 alarm.

Post-Check Verification

```
Command: show chassis hardware, Expected Output: Chassis hardware information, including temperature, voltage, and fan status, Description: This command verifies that the chassis hardware is functioning correctly and that the temperature, voltage, and fan status are within normal ranges.

Command: show chassis fpc, Expected Output: FPC status, including FPC number, state, and reason for failure, Description: This command verifies that the FPCs are functioning correctly and that there are no issues with the FPCs or line cards.

Command: show log messages | match ALARM_2730, Expected Output: Log messages related to the ALARM_2730 alarm, including timestamp and description, Description: This command verifies that the ALARM_2730 alarm has been cleared and that there are no further logs related to the issue.
```

Roll-back Procedures

- Step: Restore the original line card or ASIC, Command: request chassis fpc replace with , Description: This command restores the original line card or ASIC that was replaced, reverting the system to its original state.

- Step: Restore the original power supply, Command: request chassis psu replace with , Description: This command restores the original power supply that was replaced, reverting the system to its original state.